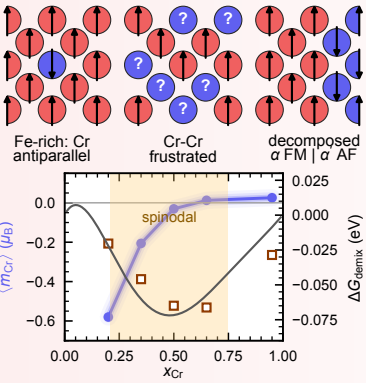
FRUSTRATE

(a) atoms & moments

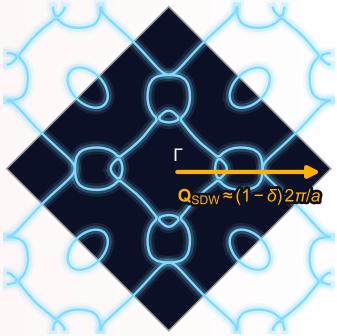
schematic



NEST

(b) Cr Fermi surface

Cr: jack (Γ) nests onto octahedron (H)
⇒ spin- and charge-density waves



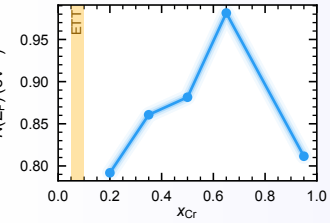
TOPOLOGY

(c) Fe-rich: ETT

ETT (Lifshitz): adding Cr moves E_F schematic

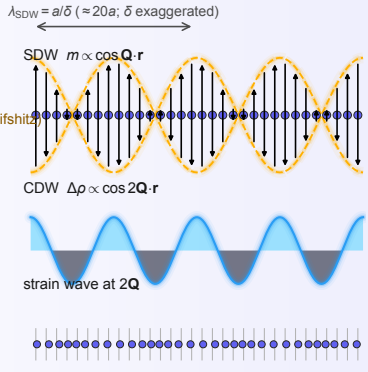


Fe↑ sheet at 5–10% Cr ⇒ √-type cusp in B, C' (Lifshitz)



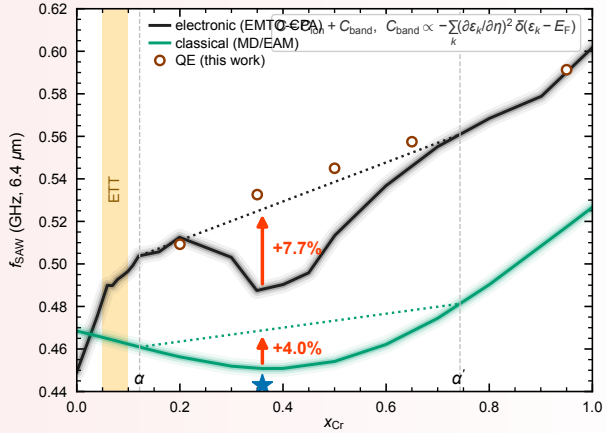
MODULATE

(d) spin & charge waves



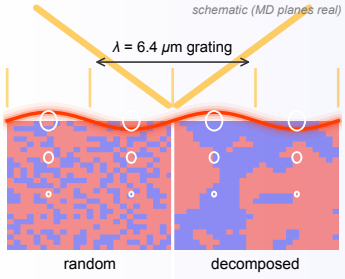
STIFFEN

(e) electron → lattice: decomposition



PROPAGATE

(f) Rayleigh wave in TGS



$C_{ij}\uparrow \Rightarrow G\uparrow \Rightarrow v_R = \eta(\nu)\sqrt{G/\rho}\uparrow \Rightarrow f = v_R/\lambda\uparrow$
Fe–36Cr, MD: 451 → 474 MHz (+5.1%)
electronic lever rule: +7.7%

DETECT

(g) (P)SAW vs pristine CASS

